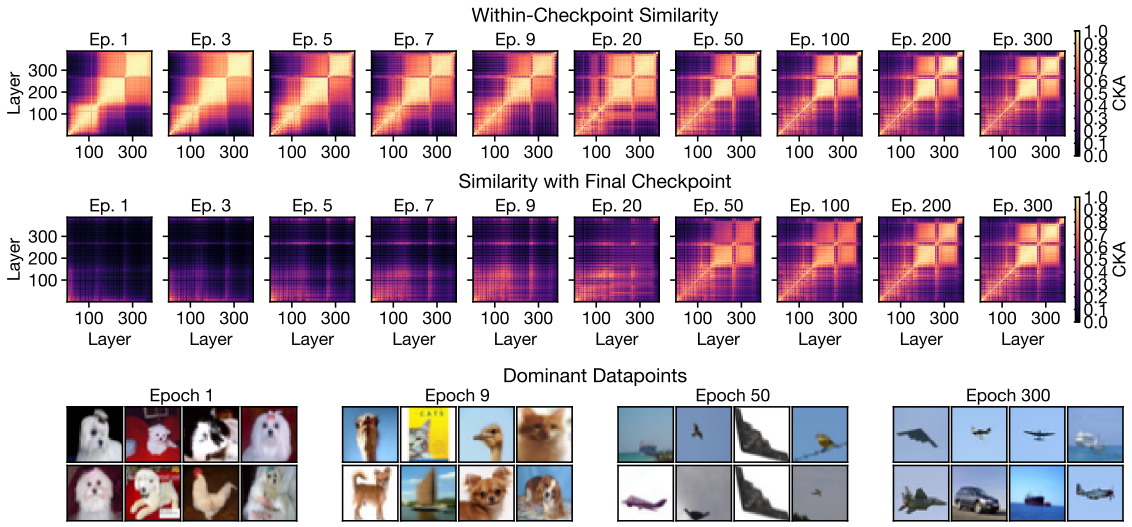

Figure 8: Block structure phenomenon arises early during training, but the corresponding dominant datapoints continue to change substantially. We compute the CKA between all pairs of layers within a ResNet-110 (1×) model at different stages of training, and find that the shape of the block structure is defined early in training (top row). However, comparing these different model checkpoints to the fully-trained model reveals that the block structure representations at different epochs are considerably dissimilar to the final representations, especially during the first half of the training process (middle row). The corresponding dominant datapoints also vary significantly over the course of training, even after the block structure is clearly visible in the heatmaps (bottom row). See Appendix F for similar plots with greater granularity, different seeds and architectures.

6 Block Structure and Training Mechanisms

Having observed how the internal representation structures — specifically, the dominant PCs of layer representations — could vary significantly during training, we turn to examining the interplay between the block structure and the training mechanism. Although the block structure arises naturally with standard training, previous work has suggested that the block structure may be an indication of redundant modules in the corresponding networks [29]. Thus, it is natural to ask whether it is possible to train large-capacity models without a block structure, and how such models perform compared to those with block structures.

Since the block structure reflects the similarity of a dominant first principal component, propagated across a wide range of hidden layers (see Section 3), we study whether regularizing the first principal components of layer activations would eliminate the block structure. More specifically, we estimate the fraction of variance explained by the first principal component of each layer using power iteration and penalize it in the training objective when it exceeds 20%. We provide full implementation details in Appendix G. The resulting heatmap, in Figure 9 top right, shows that not only does this eliminate the block structure from the internal representations, but surprisingly there is also no detrimental effect on performance (Appendix Table G.1). We even observe small accuracy improvements on CIFAR-100 and in the low-data regime, as shown in Appendix Table G.1.

Other standard training practices that are commonly used to boost performance are also effective at reducing or eliminating the block structure effect. Shake-Shake regularization [7] eliminates the

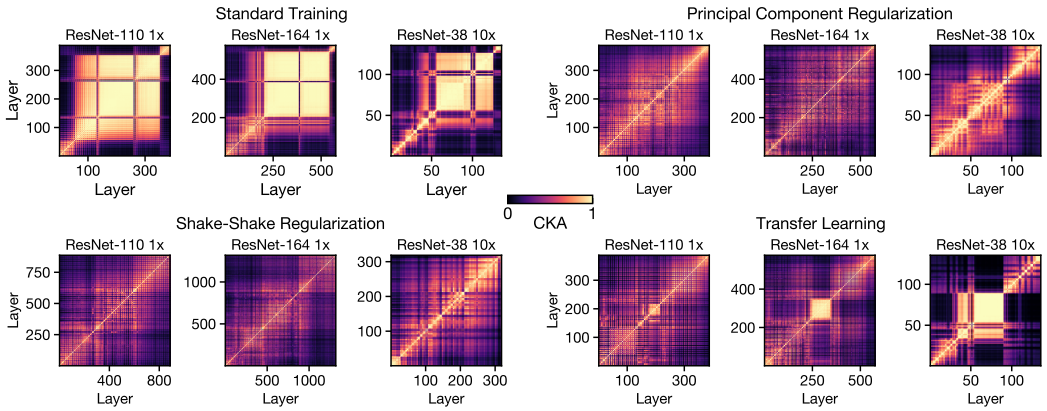

Figure 9: Training with principal component regularization, transfer learning and Shake-Shake regularization helps eliminate the block structure. We directly regularize the first PC of each layer activations given that this component explains a large fraction of variance in block structure representations, and find that this eliminates the block structure. Full algorithm details can be found in Appendix G. Shake-Shake regularization [7] has a similar effect. We also find that transfer learning reduces the appearance of the block structure, although it is still present in the largest network. These results demonstrate that the block structure phenomenon is dependent on the training mechanism. See Appendix I for implications of these training methods on representations across random seeds.

block structure for all of the network sizes that we examine (Figure 9, bottom left), whereas transfer learning (Figure 9, bottom right) and training with smaller batch sizes (Appendix H) generally reduce the appearance of the block structure, although blocks are still discernible in the largest models that we trained. In addition to regularizing the block structure, these training methods also produce more similar representations across different training runs of the same architecture configuration (Appendix Figure I.1).

Overall, our findings suggest that it is possible to obtain good generalization accuracy in networks with and without block structure. We observe that learning processes that reduce the dominance of the first PC of the representations provide slightly higher accuracies than standard training. However, some caution is warranted in interpreting these performance benefits: it may be difficult to causally determine the connection between the block structure and performance, as any training intervention targeting the block structure may have other distinct ramifications that also affect performance.

7 Scope and Limitations

Our work primarily focuses on the behavior of large-capacity networks trained on relatively small datasets. This is motivated by domains such as medical imaging where data is expensive relative to the cost of training a large model, and the high-stakes nature makes it important to understand the model's behavior. In general, understanding how the representational properties change with model capacity (relative to dataset size) is of both scientific and practical interest, as model size continues to grow over the years but there are many domains beyond vision (e.g. see Kaggle) where dataset size does not. Additional exploration is needed to study state-of-the-art settings in e.g. NLP, which use much bigger and heterogeneous datasets.

8 Conclusion

The block structure phenomenon uncovered in previous work [29] reveals significant differences in the representational structures of overparameterized neural networks and shallower/narrower ones. However, it also exhibits some contradicting behaviors — being unique to each network while propagating a dominant PC across a wide range of layers — that respectively suggest the underlying representations could either overfit to noise artifacts or capture relevant signals in the data. Our work seeks to provide an explanation for this discrepancy. We find that despite the inconsistency of the block structure across different training runs, it arises not from noise, but real and simple dataset statistics such as background color. We further discover a small set of dominant datapoints (with large activation norms) that are responsible for the block structure. These datapoints emerge early in training and vary across epochs, as well as across random seeds. We show how different training procedures, including color augmentation, transfer learning, Shake-Shake regularization, and a novel principal component regularizer, can reduce the influence of these dominant datapoints, eliminating the block structure and leading to more consistent representations across training runs.

Since the block structure phenomenon has been shown to robustly arise in large-capacity networks under various settings [29], rigorously characterizing its cause and effects is of great importance to understanding the nuances in the way these networks learn, despite their similarly good performances. We believe that insights into the representational properties of overparameterized models can benefit techniques that make direct use of the internal representations, such as transfer learning and interpretability methods. This work also motivates interesting open questions including exploring how dominant datapoints are manifested in other domains and applications of deep learning, as well as applying principal component regularization to distribution shift and self-supervision problems.

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